

Hiroataka Ito

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Education

PhD	Universitat Pompeu Fabra , Economics Advisor: Professor Geert Mesters	2022–2027 (expected)
MRes	Universitat Pompeu Fabra , Economics	2021–2022
MS	Barcelona School of Economics , Economics	2020–2021

Research Visits

Brown University , Visiting Research Fellow Host: Professor Toru Kitagawa	2024
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Research Interests

Primary field: Econometric theory

Secondary field: Applied econometrics

Topics: Partial identification, Bayesian inference, Nonparametric inference, Statistical decision theory

Job Market Paper

Asymptotic Minimax Optimality of Set Estimation

Abstract: Parameters of interest are often expressed as transformations of partially identified raw parameters. In such settings, the identified set of the targeted objects can be non-convex and there are no theoretical results on how well one can learn it from data. This paper fills part of this gap by studying the asymptotic efficiency of set estimation for a general class of identified sets. I first derive the asymptotic minimax efficiency bound for set estimators under risk functions based on the Hausdorff metric. A key ingredient in the analysis is a limit experiment framework tailored to the local structure of the identified set. I then show that a set estimator based on a criterion function is efficient under mild conditions. The analysis requires deriving the limiting distribution of the set estimator in the Hausdorff metric, which I obtain without globally characterizing the identified set via its support functions. Based on the limiting distribution, I propose a bootstrap-based confidence set. I illustrate my framework with an application to optimal taxation with behavioral agents.

Working Papers

Uniform and Locally Adaptive Quasi-Bayesian Inference for Conditional Moment Models

Teaching Experience

Barcelona School of Economics

Advanced Econometric Methods I
Master's Thesis

Graduate (PhD)
Graduate (Master)

TA
TA

2022, 2023
2022

Universitat Pompeu Fabra

Econometrics 1	Undergraduate	TA	2023, 2026
Econometrics 2	Undergraduate	TA	2022, 2023
Econometrics 3	Undergraduate	TA	2026
Introduction to Economics	Undergraduate	TA	2024

Conferences and Seminars

2024	BSE Summer Forum: Macroeconometrics and Policy Evaluation (Poster)
2023	BSE Summer Forum: Macroeconometrics and Policy Evaluation (Poster)
2022	BSE Summer Forum: Advances in Econometrics (Poster)

Awards and Scholarships

Barcelona School of Economics: Tuition Waiver	2020-2021
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Skills

Languages: English (fluent), Japanese (native), French (basic)

Programming: R, MATLAB, Python, Stata